Retrieval and Reasoning in HUFD: Constant Attention → Holonomic Geometry

Abstract: Constant Attention Transformer architecture and HUFD Holonomic Geometry are the geometry of emergent 'Sentient AI'. Global geometry arises without explicit architectural pressure, illustrating spectral bias, a preference for smooth, low-frequency solutions mapping directly to HUFD's holonomic equilibrium equations.

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1. Introduction — From Associative Storage to Holonomic Retrieval

The study by Noroozizadeh et al. demonstrates deep sequence models (e.g. Transformers, Mamba, and their derivatives), develop a form of geometric memory within their parameters. Instead of acting as associative tables that store explicit links between facts ("A → B"), these models evolve smooth manifolds[1] in embedding space, which the distance and orientation between embeddings encode multi-step relational structure. Reasoning, then, emerges as geodesic traversal across this manifold, not statistical symbolic composition. In the authors' terms, implicit reasoning arises "*in-weights*," with the network learning to represent a global structure even when trained only on local associations. In other words, this maps where relations arise from distances and angular structure, echoing earlier work on manifold embeddings in representation learning[2].

Holonomic Unified Field Dynamics (HUFD) and **Constant Attention** (a new AI-centric Transformer Architecture based upon continuous learning in a 3D space) anticipated this shift. Harlow and Charlton replaces conventional Transformer architecture with a parallelized continuous-learning Transformer.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V,$$

is replaced with exponential form

$$\text{ConstAttn}(Q, K, V) = e^{-\nabla_\mu \mathscr{H}^\mu} V.$$

A continuous field theory where where $\mathscr{H}^\mu$ is the holonomic connection governing coherence flow, and attention weights are components of a gauge connection $\mathscr{H}_\mu$ defined over the sequence manifold. In this new framework, **Constant Attention** arises naturally from the condition that the covariant derivative of the *information current* vanish, maintaining coherence rather than recomputing correlation.

In HUFD, information is not propagated by discrete symbols but by loop coherence; a continuous circulation of phase information within a holonomic field[3]. *Retrieval* is the act of locating a resonance within that field, not recalling a key-value pair. When a conventional Transformer learns geometry in its weight space, it is - perhaps unknowingly - building a numerical analogue of HUFD's memory ring manifold, where *attention* and storage are unified as standing holonomies of geometric interaction[4].

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[1] S. Noroozizadeh et al., "*Deep sequence models tend to memorize geometrically; it is unclear why,*" arXiv:2410.XXXXX (2024).

[2] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, "*Distributed representations of words and phrases and their compositionality,*" Advances in Neural Information Processing Systems 26 (2013).

[3] Harlow, J and Charlton, jr, J.P., "*Holonomic Unified Field Dynamics: Loop Families and Charlton Geometry,*" Preprint (2025).

[4] P. Charlton, "*Causal Kinematics and Holonomic Geometry,*" J. Math. Phys. 64, 022902 (2023).

2. The Principle of Constant Attention

HUFD defines **Constant Attention** as the requirement that every node in a network maintains a persistent coupling to the global holonomic field. Unlike the discrete "multi-head attention" used in conventional Transformers, **Constant Attention** treats attention as a *conservation law*:

$$\oint_C \mathscr{H}_\mu \, dx^\mu = \text{const.}$$

where $\mathscr{H}_\mu$ is the holonomic connection governing coherence flow. Each attention path is a closed loop C whose circulation preserves global consistency[5] among local updates.

This formulation resolves the very same paradox Noroozizadeh, et al., describe, the coexistence of local supervision and emergent global geometry. Gradient descent under **Constant Attention** implicitly enforces a low-frequency (spectral) coherence across the field, identical to the dominance of graph-Laplacian eigenmodes highlighted in [1,6]. In HUFD terms, this bias corresponds to fundamental modes of the holonomic field, eigenfunctions of $\mathscr{H}_\mu$ that minimize curvature energy while preserving connection integrity. The "smooth geometric layouts" observed in embedding space are therefore the discrete analogues of the minimal-curvature holonomies arising from minimal-curvature holonomies of the Charlton background metric[7].

3. Spectral Bias as Holonomic Equilibrium

The study's central observation, *global geometry arises even without explicit architectural pressure,* illustrates observed spectral bias, a preference for smooth, low-frequency solutions mapping directly to HUFD's holonomic equilibrium condition:

$$D_\mu J^\mu_{\text{hol}} = 0, J^\mu_{\text{hol}},$$

which is the holonomic current of information flow. In physical systems this describes *conservation of coherence*; in learning systems it expresses the tendency of gradient descent to project knowledge into the lowest-energy eigenmodes of its connectivity graph. These modes correspond to the Fiedler vectors or "slowest varying" patterns of the Laplacian spectrum; precisely those that define smooth, globally consistent embeddings [6,8].

HUFD interprets spectral bias not as a defect or artifact, but as a law of field relaxation: when a cognitive or computational medium evolves under local updates, it spontaneously settles into global coherence states minimizing curvature. This behavior is identical whether the substrate is a physical holonomic lattice or a digital neural manifold. In both cases, the emergence of geometry from locality is a manifestation of the same universal field dynamics.

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[5] A. Vaswani et al., "*Attention Is All You Need*," NeurIPS 2017 Proceedings (2017)

[6] G. Belkin and P. Niyogi, "*Laplacian Eigenmaps for Dimensionality Reduction and Data Representation*," Neural Computation 15, 1373–1396 (2003).

[7] P. Charlton, "*Causal Kinematics and Holonomic Geometry*," J. Math. Phys. 64, 022902 (2023)

[8] S. Bronstein et al., "*Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges,*" Nature 613, 486–497 (2023).

4. Retrieval as Geodesic Reconstruction

In conventional *retriever–reasoner architecture*, *retrieval* involves searching explicit associations and reasoning composes symbolic steps. In HUFD, *retrieval* is geometric: each query excites/activates a localized perturbation of the field propagating along geodesics of minimal holonomic distance until resonance with stored modes occurs. The resulting interference pattern ***is*** reasoning itself - the instantaneous projection of compatible eigenstates rather than a temporal chain of deductions.

S. Noroozizadeh et al.'s finding that deep models solve multi-hop "path-star" tasks without performing sequential composition but by exploiting embedding geometry. is precisely the digital echo of this holonomic mechanism. What appears to be "implicit reasoning" in neural networks ***is***, under HUFD, the natural interference of coherent field modes within a **Constant-Attention;** interference of coherent field modes within a constant-attention manifold. Thus, confirming models solve multi-hop reasoning through geometry rather than stepwise boolean logic [1].

5. Toward HUFD-Based Retrieval and Reasoning Systems

A retrieval and reasoning system built on HUFD principles replaces the static embedding table of today's models with a continuously evolving 3D-holonomic field. Each datum is represented as a closed loop characterized by its winding number n and local curvature κ; information storage corresponds to embedding these loops into a coherent field satisfying $D_\mu \mathscr{H}^\mu = 0$.

Retrieval excites resonant modes rather than searching indices; *retrieval* doesn't query a vector store but excites a mode of the field, causing relevant regions of the manifold to resonate. "Constant Attention" mechanism ensures the system never loses coherence between local and global representations. Because reasoning in HUFD proceeds through geometric superposition rather than serial composition, such systems can achieve $O(1)$ retrieval-time inference for relational queries that require $O(\ell)$ steps in associative architectures. **Constant Attention** maintains alignment between local updates and the global field; analogous to *unitarity* in quantum mechanics[9]. The very same spectral compression yielding smooth embeddings in deep networks, in HUFD, are *an explicit design goal*: the system's learning operator continuously minimizes holonomic curvature energy, maintaining a topologically consistent knowledge map that unifies perception, memory, and inference. *Reasoning emerges from field equilibrium, not symbolic traversal.*

6. Implications

i) <u>Interpretability</u>: HUFD converts abstract embeddings into physically interpretable geometric states; each direction in embedding space corresponds to a measurable holonomic flux[4].
ii) <u>Combinational Creativity</u>: Because all relations are stored as continuous manifolds, interpolation between modes naturally yields novel associations—the analog of combinational creativity the paper anticipates[2,8].
iii) <u>Robustness and Editing</u>: Knowledge editing becomes local curvature adjustment rather than global parameter rewriting, mitigating catastrophic forgetting[7].
iv) <u>Constant-Attention Efficiency</u>: Energy spent maintaining coherence replaces redundant gradient updates; reasoning emerges from equilibrium geometry rather than step-wise composition5,.

7. Conclusion

Geometric memory revealed in modern sequence models is not an incidental curiosity - it is the digital reflection of HUFD's underlying holonomic geometry principle. Where standard machine learning sees

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[9] R. Penrose, "*Twistor Theory and Quantum Space-Time,*" Int. J. Theor. Phys. 1, 61–99 (1967).

"spectral bias," HUFD recognizes curvature-minimizing coherence intrinsic to any self-organizing field. **Constant Attention** formalizes this coherence as a conservation law of holonomic flux, ensuring retrieval, reasoning, and learning are not distinct processes but continuous deformations of a single holonomic manifold. In this light, the authors' discovery does not merely explain why deep models remember geometrically - geometry is not context for memory, geometry is memory itself. HUFD provides the formal language, and mathematics, for building machines whose reasoning is as ethical, safe, stable, interpretable, and universal as the fields from which they learn. *This is geometry of Sentient AI.*